

Python For Data Analytics Assignment-2

1.Loading data:

```
#Loading the Taxis Dataset using the following code:
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = sns.load_dataset("taxis")
df.head()
```

fare	tip	tolls	total	color	payment	pickup_zone	dropoff_zone	pickup_borough	dropoff_borough
7.0	2.15	0.0	12.95	yellow	credit card	Lenox Hill West	UN/Turtle Bay South	Manhattan	Manhattan
5.0	0.00	0.0	9.30	yellow	cash	Upper West Side South	Upper West Side South	Manhattan	Manhattan
7.5	2.36	0.0	14.16	yellow	credit card	Alphabet City	West Village	Manhattan	Manhattan
27.0	6.15	0.0	36.95	yellow	credit card	Hudson Sq	Yorkville West	Manhattan	Manhattan

2.Handled Missing Value:

```
#Handle The Missing Values
print(df.isnull().sum())

# Categorical -> mode
for col in ["payment", "pickup_zone", "dropoff_zone",
            "pickup_borough", "dropoff_borough"]:
    df[col] = df[col].fillna(df[col].mode()[0])

# Numerical -> median (only if any are missing)
for col in df.select_dtypes("number").columns:
    df[col] = df[col].fillna(df[col].median())

# Critical columns that can't be reasonably imputed -> drop rows
df = df.dropna(subset=["pickup", "dropoff", "fare", "distance"])

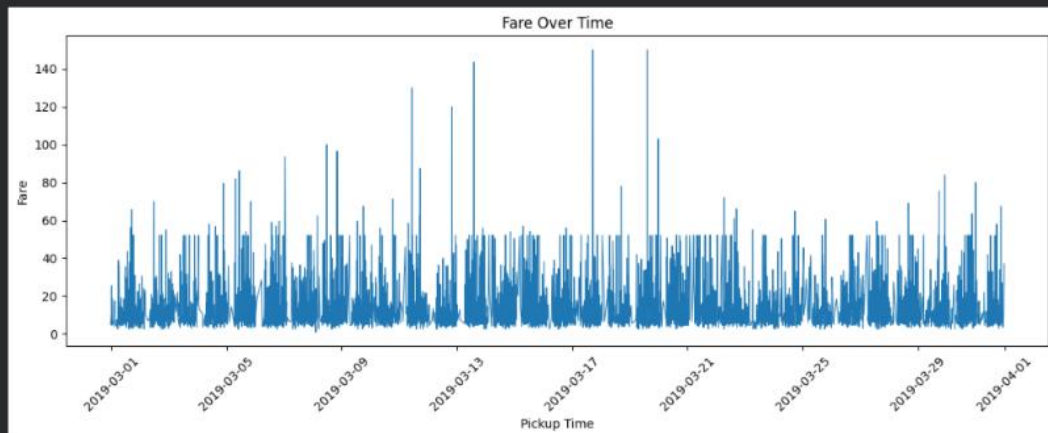
print(df.isnull().sum().sum(), "missing values left")
```

pickup	0
dropoff	0
passengers	0
distance	0
fare	0

3. Visualizations using Matplotlib/Pandas Plot:

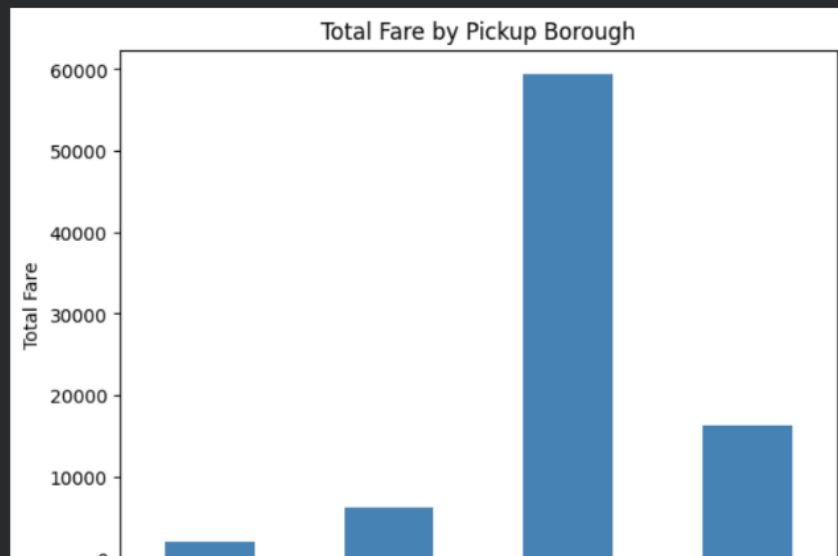
➤ Line Chart:

```
# Line chart: fare over time
df["pickup"] = pd.to_datetime(df["pickup"])
d = df.sort_values("pickup")
plt.figure(figsize=(12, 5))
plt.plot(d["pickup"], d["fare"], linewidth=0.8)
plt.title("Fare Over Time"); plt.xlabel("Pickup Time"); plt.ylabel("Fare")
plt.xticks(rotation=45); plt.tight_layout(); plt.show()
```



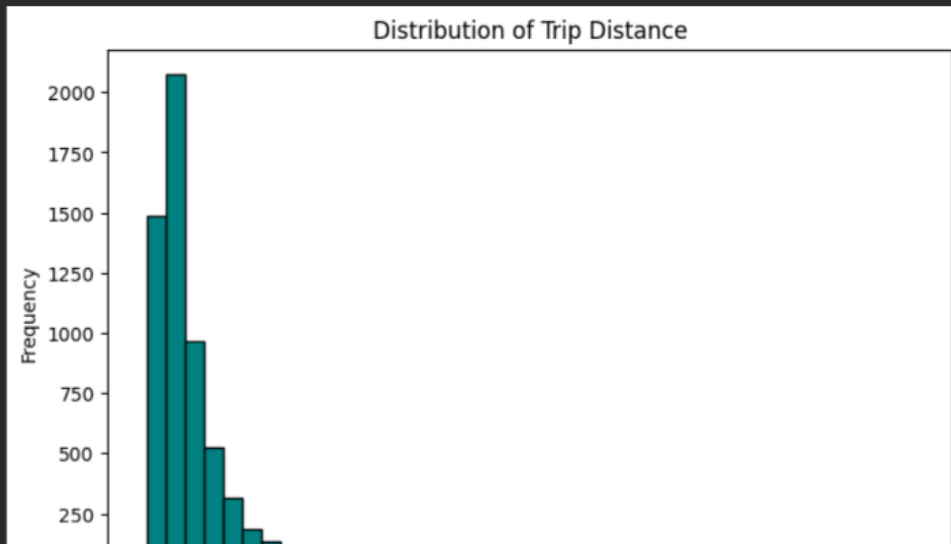
➤ Bar Chart:

```
# Bar chart: total fare per pickup borough
df.groupby("pickup_borough")["fare"].sum().plot(kind="bar", color="steelblue", figsize=(7, 5))
plt.title("Total Fare by Pickup Borough"); plt.ylabel("Total Fare")
plt.xticks(rotation=0); plt.show()
```



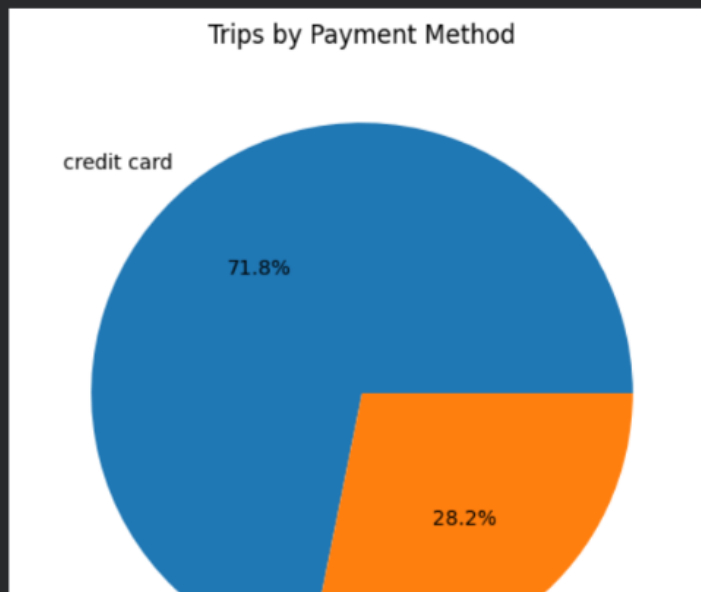
➤ Histogram:

```
# Histogram: distance
plt.figure(figsize=(8, 5))
plt.hist(df["distance"], bins=40, color="teal", edgecolor="black")
plt.title("Distribution of Trip Distance"); plt.xlabel("Distance (miles)"); plt.ylabel("Frequency")
plt.show()
```

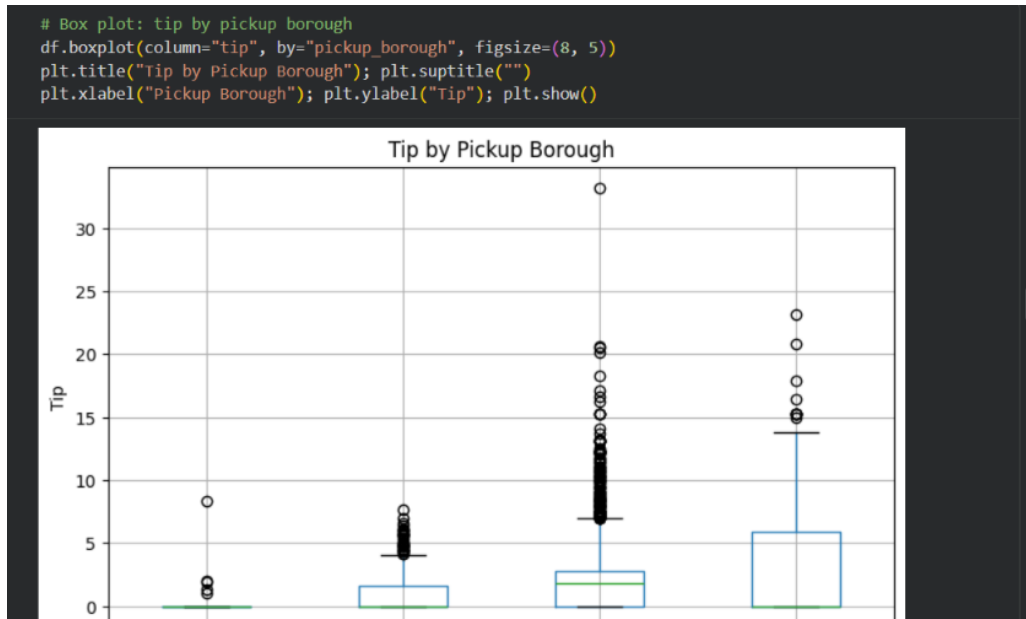


➤ Pie Chart:

```
# Pie chart: trips by payment method
df["payment"].value_counts().plot(kind="pie", autopct="%1.1f%%", figsize=(6, 6))
plt.title("Trips by Payment Method"); plt.ylabel(""); plt.show()
```

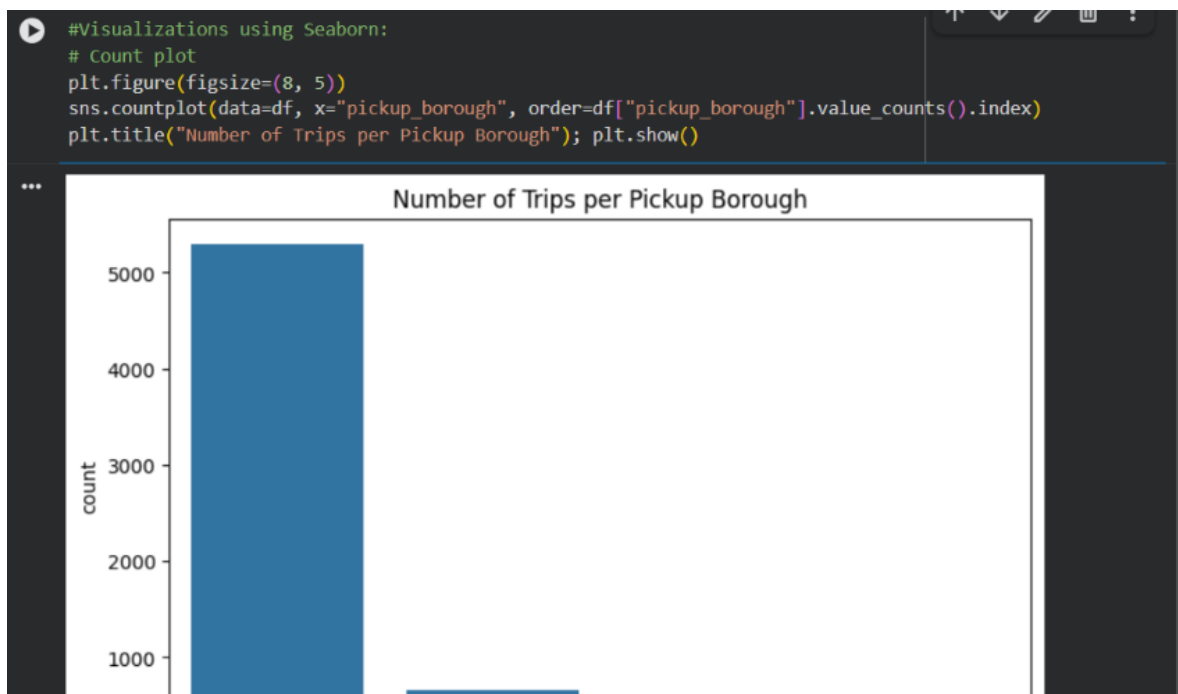


➤ Box Plot:

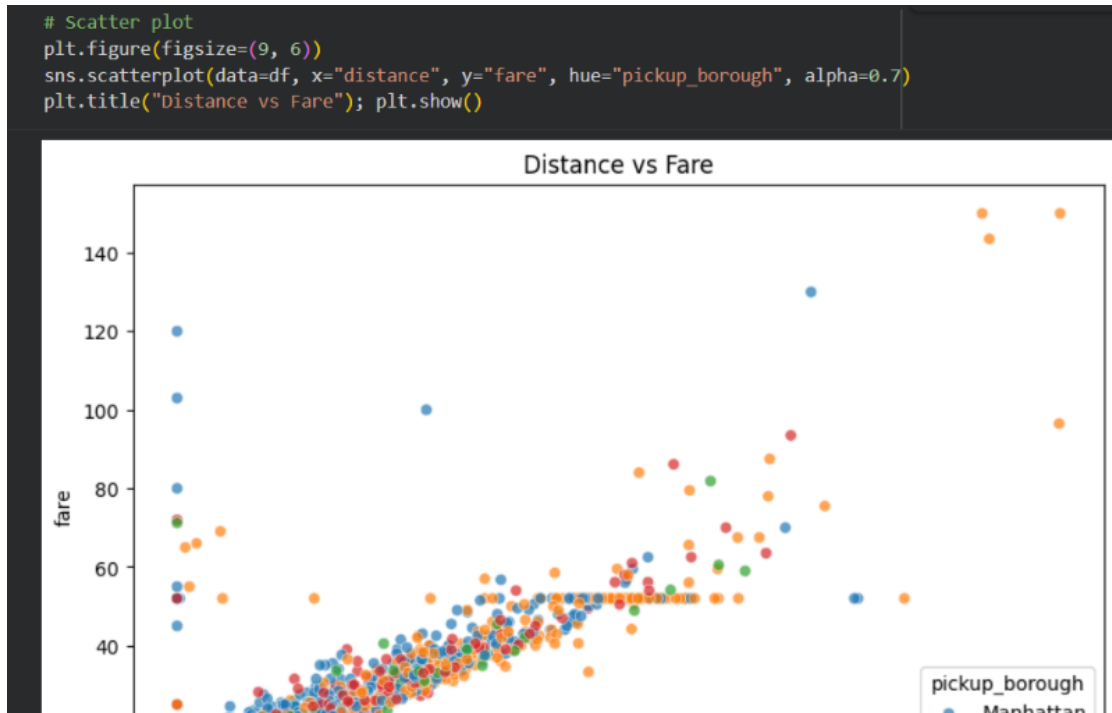


Visualizations using Seaborn:

➤ Count Plot:



➤ Scatter Plot:

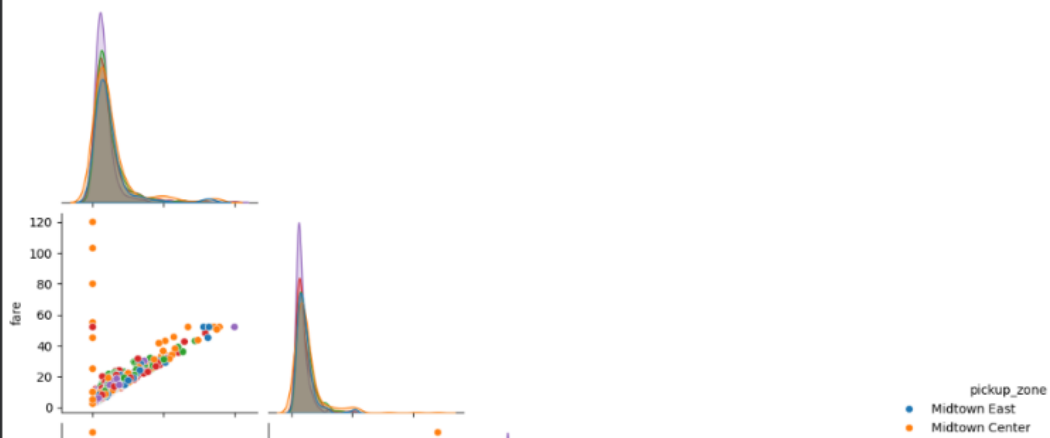


➤ Heatmap:



➤ Pair Plot:

```
# Pair plot (pickup_zone has ~190 zones, so colour only the top 5)
top_zones = df["pickup_zone"].value_counts().head(5).index
sub = df[df["pickup_zone"].isin(top_zones)]
sns.pairplot(sub, vars=["distance", "fare", "tip", "total"], hue="pickup_zone", corner=True)
plt.show()
```



➤ Violin Plot:

```
# Violin plot
plt.figure(figsize=(8, 5))
sns.violinplot(data=df, x="payment", y="fare")
plt.title("Fare Distribution by Payment Method"); plt.show()
```

